Plant Growth and Development

1. **"A plant after decapitation becomes bushy". The possible reason of above phenomenon is**
(1) Removal of apical buds promote growth of lateral buds into branches
(2) Removal of apical buds induce stem elongation as auxin suppresses production of gibberellins
(3) Apical buds release some hormone which kills lateral buds
(4) Lateral buds have auxins which are suppressed in presence of cytokinin

2. **Given below are two statements: One is labelled as Assertion (A) and other is labelled as Reason (R).**
Assertion (A): Abscisic acid is known as stress hormone.
Reason (R): Synthesis of ABA is stimulated by drought, water logging and other adverse environmental conditions and it increases the tolerance of plants towards it.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

3. **Plants follows different pathways in response to environment or phases of life to form different kinds of structures this ability is called :**
(1) Plasticity
(2) Maturity
(3) Elasticity
(4) Flexibility

4. **Select the incorrectly matched pair.**
(1) Auxins – stimulates shoot formation on stem cuttings
(2) Cytokinin – Delay the senescence of intact leaves
(3) Gibberellins – Increase the length of grapes stalks
(4) Ethylene – Accelerates shedding of leaves, flowers and fruits

5. **Removal of shoot tips is a very useful technique to boost the production of tea leaves this is because :**
(1) Gibberellins present bolting and are inactivated
(2) Auxins prevent leaf drop at early stages
(3) Effect of auxins is removed and growth of lateral buds is enhanced
(4) Gibberellins delay senescence of leaves

6. **If takes very long time for pineapple plants to produce flowers which combination of hormones can be applied to artificially induce flowering in pineapple plants throughout the year to increase yield :**
(1) Cytokinin and abscisic acid
(2) Auxin and ethylene
(3) Gibberellin and cytokinin
(4) Gibberellin and abscisic acid

7. **You are given a tissue with its potential for differentiation in an artificial culture which of the following pairs of hormones would you add to the medium to secure shoots as well as roots ?**
(1) IAA and Gibberellin
(2) Auxin and cytokinin
(3) Auxin and abscisic acid
(4) Gibberellin and abscisic acid

8. **Read the following Assertion (A) and Reason (R)**

Assertion (R) : Several hormonal and structural changes are initiated which lead to the differentiation and further development of the floral primordium.

Reason (A) : Shoot apical meristem transforms into floral meristem

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

9. **Select the correct option for the hormone which was first isolated from human urine.**

(1) Widely applied in tea-plantations for hedge-making

(2) Controls xylem differentiation and helps in cell division

(3) Acidic and is used to increase the length of grapes stalks

(4) Induces seed and bud dormancy

10. **If the apical bud has been removed then we observe :**

(1) More lateral branches

(2) More axillary buds

(3) Plant growth stops

(4) Flowering stops

11. **Who coined the term kinetin:**

(1) Skoog and miller

(2) Cousin

(3) F.W. Went

(4) Kurosawa

12. **Vernalisation stimulates flowering in :**

(1) Turmeric

(2) Carrot

(3) Ginger

(4) Zaminkand

13. **Match the column-I with column-II**

	Column-I		Column-II
a.	E. Kurosawa	i.	He confirmed the release of a volatile substance from ripened oranges
b.	Miller et al.	ii.	He reported the appearance of symptoms of the foolish seedling in rice when they were treated with sterile filtrate fungus.
c.	F.W. Went	iii.	Identified crystallised and the cytokinesis promoting active substance
d.	H.H. Cousins	iv.	Isolated an indole compound from tips of coleoptiles of oat seedlings

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(iv), c(iii), d(i)

14. **Cork, secondary cortex and secondary xylem are formed through**

(1) Dedifferentiation

(2) Redifferentiation

(3) Differentiation

(4) Obliteration

15. **At day one, the area of leaf A was 10 cm2 and that of leaf B was 100 cm2.On next day,area of leaf B was 110 cm2 and that of leaf A was 20 cm2 . The relative growth rate of leaf B and A is**

(1) 20% and 19%

(2) 100% and 10%

(3) 10% and 100%

(4) 11% and 20%

16. Buttercup and coriander differ from each other in terms of development as in the former

(1) The plant follow different pathway in response to phases of life to form different kinds of structures

(2) The leaves of the juvenile plants are different in shape from those of mature plants.

(3) The sequence of the development process ultimately lead to senescence and then death

(4) There is difference in shape of leaves produced in air with those produced in water

17. Which of the following statements are correct regarding growth?

(A) In plants, form of growth is open and localised

(B) Swelling of piece of wood in water is considered as growth since it involve the increase in size

(C) Growth is accompanied by metabolic processes

(D) Growth, at a cellular level, is a result of increase in the amount of protoplasm

(1) All the statements are correct

(2) A and B

(3) B, C and D

(4) A, C and D

18. What is the site of perception of photoperiod necessary for induction of flowering in plants :

(1) Leaves

(2) Lateral buds

(3) Pulvinus

(4) Shoot apex.

19. Removal of shoot tips is a very useful technique to boost the production of tea leaves. This is because: -

(1) Gibberellins prevent bolting and are inactivated

(2) Auxins prevent leaf drop at early stages

(3) Effect of auxins is removed and growth of lateral buds is enhanced.

(4) Gibberellins delay senescence of leaves

20. One of the commonly used plant growth hormone in tea plantations is - :

(1) Abscisic acid

(2) Zeatin

(3) Indole -3- acetic acid

(4) Ethylene

21. Given below are two statements :

Assertion: Auxin is used in plant propagation widely.

Reason: Auxin initiate rooting.

Choose the correct option –

(1) Both Assertion and Reason are correct, and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are correct, but Reason doesn't explain Assertion

(3) Assertion is correct but Reason is wrong

(4) Both Assertion and Reason are incorrect

22. The three phases of growth in correct order is-

(1) Meristematic, maturation, elongation

(2) Elongation, meristematic, maturation

(3) Meristematic, elongation, maturation

(4) Elongation, maturation, meristematic

23. Given below are two statements:

Statement-I : Plant growth is unique because plants retain the capacity for unlimited growth throughout their life.

Statement-II : Growth, at a cellular level, is principally a consequence of increase in the amount of protoplasm.

In the light of the above statements, choose the most appropriate answer from option given below

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

24. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

Assertion (A) : Development is a term that includes all changes that an organism goes through during its life cycle.

Reason (R) : Plants follow different pathways in response to environment or phases of life to form different kinds of structure.

Choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

25. Apical dominance can be counteracted by

(1) Application of ethylene

(2) Application of BAP

(3) Removal of lateral buds

(4) Providing auxins to plant

26. Match the following :

(A) Human urine	**(i) Ethylene**
(B) Coconut milk	**(ii) GA3**
(C) Ripened oranges	**(iii) Auxin**
(D) Gibberella fujikuroi	**(iv) Cytokinin**

(1) A - iii, B - iv, C - i, D - ii

(2) A - i, B - ii, C - iii, D - iv

(3) A - iv, B - iii, C - ii, D - i

(4) A - ii, B - i, C - iv, D – iii

27. Which of the following phenomenon in plants requires interaction of more than one plant growth regulator :-

(1) Dormancy in seeds/buds

(2) Apical dominance

(3) Senescence

(4) All of the above

28. (i) Promote nutrient mobilisation

(ii) Overcome the apical dominance

(iii) Lateral shoot growth

All the above physiological effects are related to which of the following plant growth regulators?

(1) Auxins

(2) Gibberellins

(3) Abscisic acid

(4) Cytokinins

29. The plant growth regulators (PGRs) are

(1) Simple organic substances of different chemical composition

(2) Complex organic substances of different chemical composition

(3) Simple and complex organic substances of same chemical composition

(4) Small organic substances of same chemical composition

30. Identify the correct set of statements :

(a) Cytokinins helps to promote apical dominance.

(b) Ethylene is synthesised in large amounts by tissues undergoing senescence and ripening fruit.

(c) ABA plays and important role in seed development, maturation and dormancy.

(d) Seed dormancy is when seeds fails to germinate even with external conditions are favourable.

Choose correct answer from options given below

(1) a, b and c only

(2) b, c and d only

(3) a, c and d only

(4) a, b and d only

31. Vivipary is seen in

(1) Mango and Citrus

(2) Cherry and rice

(3) Eichhornia and henbane

(4) Sonneratia and Rhizophora.

32. Match the following (Column - I with Column II)

Column I	Column II
a. Weedicide	(i) GA3
b. Bolting	(ii) Cytokinin
c. Thinning of cotton	(iii) Ethylene
d. Lateral shoot growth	(iv) 2, 4-D

(1) a(iii), b(ii), c(iv), d(i)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(iv), b(ii), c(iii), d(i)

33. Which of the following was one of the first gibberellins to be discovered and is the most intensively studied form?

(1) GA1

(2) GA2

(3) GA3

(4) GA4

34. Which of the following is a derivative of acetyl CoA and also responsible for delaying the ripening of fruits?

(1) Gibberellin

(2) Auxin

(3) Cytokinin

(4) ABA

35. Given below are two statements:

Statement-I : One single maize root apical meristem can give rise to more than 17,500 new cell per day.

Statement-II : Cells in watermelon may increase in size by upto 3,50,000 times.

Choose the most appropriate answer

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

36. "Avena" coleoptile curvature test is used for bioassay of a hormone 'X'."

'X' can be artificially used to

a. Induce parthenocarpy in tomatoes.

b. Promote shooting in callus.

c. Remove dicot weeds from crop fields.

(1) a only

(2) b only

(3) b and c

(4) a and c

37. During geometric growth of an organism, the initial phase is

(1) Lag phase

(2) Exponential phase

(3) Log phase

(4) Stationary phase

38. Plant growth as compared to animals is unique because

(1) It is accompanied by metabolic processes, that occur at the expense of energy

(2) They retain the capacity for unlimited growth throughout their life due to presence of meristems

(3) Growth in plants is at the tips only

(4) There is an irreversible permanent increase in size of an organ or its part or even of an individual cell

39. Identify the PGRs that naturally occur in plants

(a) Kinetin

(b) IAA

(c) 2, 4-D

(d) Zeatin

(e) GA3

Select the correct option

(1) (a), (b) and (d)

(2) (b), (d) and (e)

(3) (a) and (c)

(4) (a), (b), (d) and (e)

40. Development in plants is controlled by

(1) PGR only

(2) Environmental, genetic and intercellular factors

(3) Intracellular factors only

(4) Intrinsic factors only

41. Abscission promoting growth hormones are

(1) Auxin and cytokinin

(2) Ethylene and Gibberellin

(3) Cytokinin and ABA

(4) ABA and ethylene

42. The cells proximal to meristematic zone have/show

(a) Large conspicuous nuclei

(b) Thin cellulosic cell wall

(c) Increased vacuolation

(d) New cell wall deposition

(e) High rate of cell division

Select the correct ones.

(1) (a), (b) and (e) only

(2) (b), (c) and (d) only

(3) (c) and (d) only

(4) (a) and (e) only

43. Cell positioned away from root apical meristems differentiate as ___(A)___cells, while those pushed to the periphery matures as ___(B)___.

Select the correct option to fill in the blank respectively.

(1) Root-cap ; Endodermis

(2) Root hair ; Epidermis

(3) Root-cap ; Epidermis

(4) Root hair ; Endodermis

44. Dormancy of seeds is induced by hormone that also

(1) Inhibits the protein and RNA synthesis in leaves

(2) Induces fruit ripening

(3) Activates the α -amylase in seeds

(4) Induces rooting and root hair formation

45. Identify the correct set of statements :

(a) A sigmoid curve is a characteristic of living organisms growing in a natural environment.

(b) Differentiated cells can regain the capacity to divide is termed as dedifferentiation.

(c) The plant growth regulators are small simple molecules of diverse chemical composition.

(d) Auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) a, b and c only

(2) b, c and d only

(3) a, c and d only

(4) a, b, c and d

46. A phytohormone that helps the plants to increase their absorption surface by promoting root hair formation is

(1) Gibberellic acid

(2) Ethylene

(3) Zeatin

(4) Cytokinin

47. Read the following w.r.t. auxin.

i. It has been extensively used in agricultural practices.

ii. It promotes fruit and leaf drop at early stages of plant.

iii. It was first isolated from tip of coleoptile of oat.

iv. It is composed of indole compound.

The correct ones are

(1) i, iii and iv only

(2) i and iv only

(3) ii and iii only

(4) All i, ii, iii and iv

48. **Auxin helps in all, except**

(1) Apical hook formation

(2) Apical dominance

(3) Cell division

(4) Xylem differentiation

49. **All the given plant tissues are re-differentiated, except.**

(1) Secondary phloem

(2) Secondary xylem

(3) Cork

(4) Cork cambium

50. **Flowering in pineapple is promoted by**

(1) Auxin and ABA

(2) Cytokinin and ethylene

(3) Auxin and ethylene

(4) ABA and cytokinin

Answer Key

1. (1)
2. (1)
3. (1)
4. (1)
5. (3)
6. (2)
7. (2)
8. (2)
9. (2)
10. (1)
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44. (1)
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46. (2)
47. (2)
48. (1)
49. (4)
50. (3)